

Quiz 6

● Graded

Student

Samyak Sanghvi

Total Points

2.75 / 4 pts

Question 1

Q1

2 / 2 pts

✓ + 0.5 pts Linearly independent: yes

✓ + 0.5 pts for $c_0.t_0(x)+c_1.t_1(x)+c_2.t_2(x)=0$ at $x=0,1,2$ (for all x) $\Rightarrow c_0=c_1=c_2=0$

✓ + 0.5 pts Orthogonal: no

✓ + 0.5 pts integral $t_0(x) t_1(x) dx$ is not zero

+ 0 pts No/Wrong answer

Question 2

Q2

0.75 / 2 pts

✓ + 0.5 pts $s_0=\sqrt{1/N}$ =constant line

+ 0.5 pts s_1 should be decreasing from $\sqrt{2/N}$ to $-\sqrt{2/N}$

+ 1 pt $\cos(\cdot)$ has all values 1 when $u = 0$ while it has some values less than 1 when $u \neq 0$, so needs larger normalization

+ 0 pts No/Wrong answer

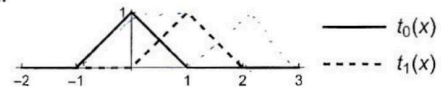
💬 + 0.25 pts Point adjustment

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COL7(6)83 Quiz 6
23 September, 2025

Entry no.: 2023CS10807

1. Consider the tent functions, $t_i(x) = \max(1 - |x - i|, 0)$, and let $S = \{t_0, t_1, t_2\}$. (i) Are the functions in S linearly independent? (ii) Are they orthogonal? Justify your answer in both cases.



(i) ~~no~~ Yes

as $t_2(2.5) > 0$ but $t_0(2.5) = 0$, $t_1(2.5) = 0$

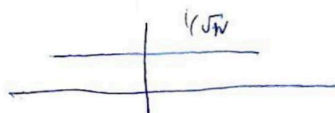
$\therefore$ for no α, β $\alpha t_0 + \beta t_1 = t_2$

(ii) No

as $t_0 \cdot t_1 \neq 0$

2. Recall that the discrete cosine transform uses basis functions $s_u(x) = \alpha_u \cos(\pi u(x + \frac{1}{2})/N)$, where $\alpha_u = \sqrt{1/N}$ if $u = 0$ and $\sqrt{2/N}$ otherwise. Sketch $s_0(x)$ and $s_1(x)$, and explain informally why a different factor is used for $u = 0$.

$s_0(x)$



$s_1(x)$

